

Learning target DF7, version 1

Consider the curve defined implicitly by the equation

$$xy^3 + 2xy = 6$$

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y . Show all of your work and thinking clearly, using correct notation.
2. Notice that the point $(2, 1)$ lies on the curve $xy^3 + 2xy = 6$. Find the exact slope of the tangent line to curve at $(2, 1)$. Show clearly how you determined this value, and label your work with proper notation.

Learning targets DF1 and DFa, version 2

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table¹ gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t .

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

1. Compute an approximation for $A'(1960)$, and **include units** for this number.
Write a sentence explaining what the number means about how the area of the glacier is changing.
Don't say "per," and don't say "rate."
2. Compute an approximation for $A'(1999)$, and **include units** for this number.
Do you think your approximation is too high or too low? Why?
3. How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

¹Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research*, 42(2), 198-209.

Learning target DF2, version 1

Suppose that $f(x) = 3x^2 - 5x + 4$.

1. Use the limit definition of the derivative to find $f'(x)$. **No shortcut rules!**

2. Evaluate at $x = 8$.

3. (Bonus!) What happens to the 3? What about the -5 ? And the 4?

Learning target DF2, version 2

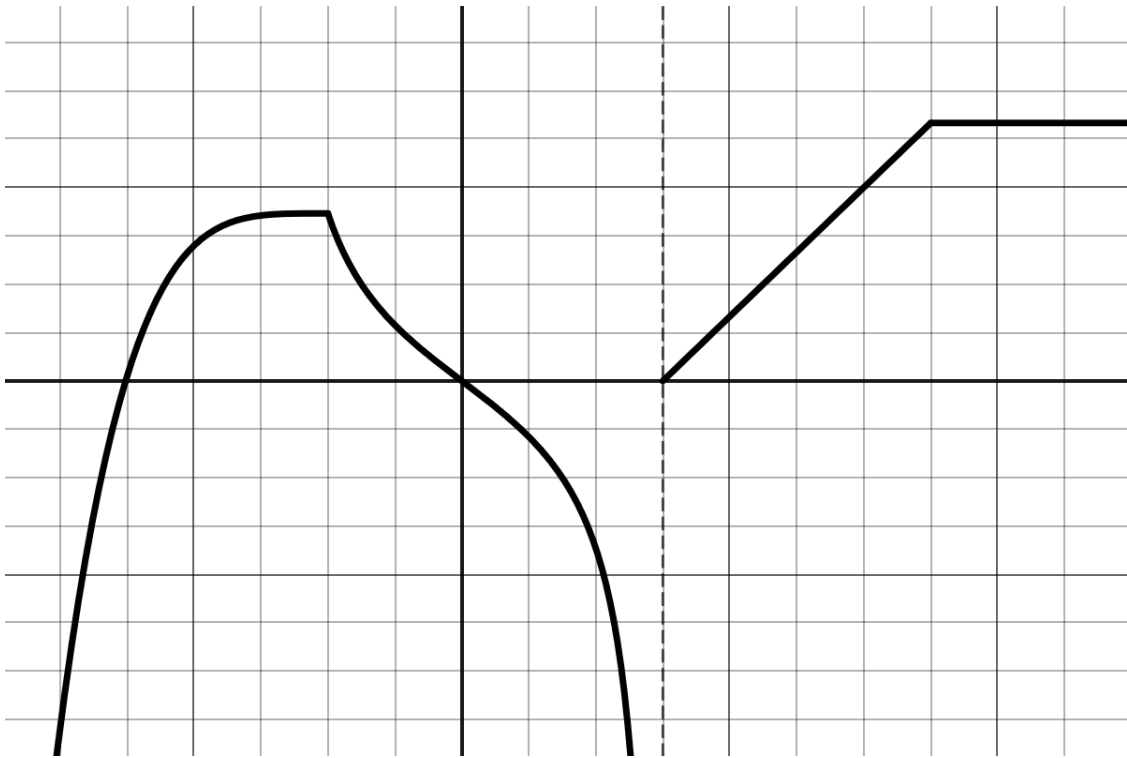
Suppose that $g(w) = 6w^3 - 2w^2 - 9w - 2$. Use the limit definition of the derivative to find $g'(w)$.

Algebra hint: $(w + h)^3 = w^3 + 3w^2h + 3wh^2 + h^3$.

PS - The answer is $g'(w) = 18w^2 - 4w + 9$.

Learning target DFb, version 2

Here is the graph of some wacky function $h(t)$:



Sketch the graph of $h'(t)$ on the blank axes below.

